

BDAT 614: Statistical Quality Control and Improvement

Statistical Methods for Process Monitoring, Control & Excellence

Prof. Louis Asiedu

May 21, 2026

Course Description & Relevance

- Comprehensive introduction to quality control using statistical methods
- Key topics: Process variability, Control Charts, Process Capability, Acceptance Sampling, Six Sigma & Lean, TQM
- Big Data Focus: Real-time SPC, IoT/sensor data, Spark streaming, ML-enhanced monitoring
- Goal: Equip students with industry-ready skills for data-driven quality improvement

Learning Objectives

- Present fundamental principles of Statistical Process Control (SPC)
- Design, construct and interpret control charts for variable and attribute data
- Evaluate process capability and performance
- Apply statistical techniques to identify and resolve quality issues
- Integrate classical SQC with modern Big Data Analytics tools

Course Overview

- Fundamental concepts of Quality, Quality Control, and Quality Assurance
- Dimensions of Quality and their importance in data-driven environments
- Introduction to Statistical Quality Control (SQC)
- Introduction to Quality and Variation
- Statistical Process Control (SPC) Fundamentals
- Control Charts: Detailed Design & Examples
- Process Capability Analysis

Course Overview

- Shewhart Cycle (PDCA/PDSA)
- DMAIC Framework
- Measurement System Analysis (Gauge R&R with ANOVA)
- Acceptance Sampling
- Six Sigma, Lean Tools & DMAIC
- Continuous Improvement and Total Quality Management (TQM)
- Big Data Applications in SQC
- Summary and Case Studies

What is Quality?

Classic Definitions

- **Fitness for Use** — Joseph Juran
- **Conformance to Requirements** — Philip Crosby
- **Delight the Customer** — Modern perspective
- **Predictable Degree of Uniformity and Dependability** — W. Edwards Deming

In Big Data Context: Quality of data, algorithms, models, and insights.

Dimensions of Quality (David Garvin)

- 1 **Performance** — Primary operating characteristics
- 2 **Features** — Secondary characteristics
- 3 **Reliability** — Probability of failure-free operation
- 4 **Conformance** — Meeting specifications
- 5 **Durability** — Useful life
- 6 **Serviceability** — Speed and ease of repair
- 7 **Aesthetics** — Look, feel, sound, taste, smell
- 8 **Perceived Quality** — Reputation and image

Note: Different stakeholders may prioritize different dimensions.

Quality Control vs Quality Assurance

Quality Control (QC)

- Reactive
- Product-oriented
- Detection of defects
- Statistical tools, inspection, testing
- "Find and fix"

Statistical Quality Control (SQC)

- Uses statistics to monitor and control processes

Quality Assurance (QA)

- Proactive
- Process-oriented
- Prevention of defects
- System design, audits, standards (ISO 9001)
- "Build quality in"

ISO 9001: Quality Management System

ISO 9001 is an internationally recognized standard for **Quality Management Systems (QMS)** developed by the International Organization for Standardization (ISO).

Current Version:

- ISO 9001:2015

Purpose of ISO 9001

- Ensure consistent product and service quality
- Enhance customer satisfaction
- Improve operational efficiency
- Promote continual improvement
- Meet regulatory and statutory requirements

ISO 9001: Quality Management System

Key Principles

- Customer focus
- Leadership
- Engagement of people
- Process approach
- Continuous improvement
- Evidence-based decision making
- Relationship management

Benefits of Certification

- Increased credibility and competitiveness
- Improved risk management
- Better process consistency
- Enhanced organizational performance

Introduction to Statistical Quality Control (SQC)

- Application of statistical methods to monitor, control, and improve quality
- Pioneered by Walter A. Shewhart (1920s)
- Key Contributors: Shewhart, Deming, Juran, Ishikawa
- Core Philosophy: **Reduce Variation**

Goal: Make processes predictable and capable of meeting customer requirements.

Process Variation – The Heart of Quality Control

"Variation is the enemy of quality." — W. Edwards Deming

All processes exhibit variation. Understanding, measuring, and reducing variation is the foundation of Statistical Quality Control.

What is Process Variation?

- Natural differences in process output even when inputs are the same
- No two products or services are ever exactly identical
- Measured as dispersion (spread) around the target value
- Sources: Raw materials, machines, methods, people, environment, measurement systems

Goal of SQC: Reduce variation to improve consistency and predictability.

Two Types of Process Variation

Common Cause Variation

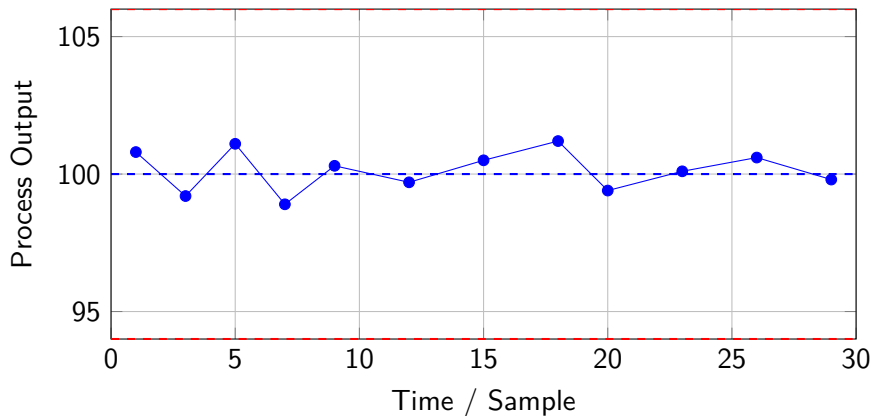
- Inherent in the process
- Random and predictable
- Many small, unavoidable causes
- Process is **stable** (in statistical control)
- Requires system-level improvement

Special Cause Variation

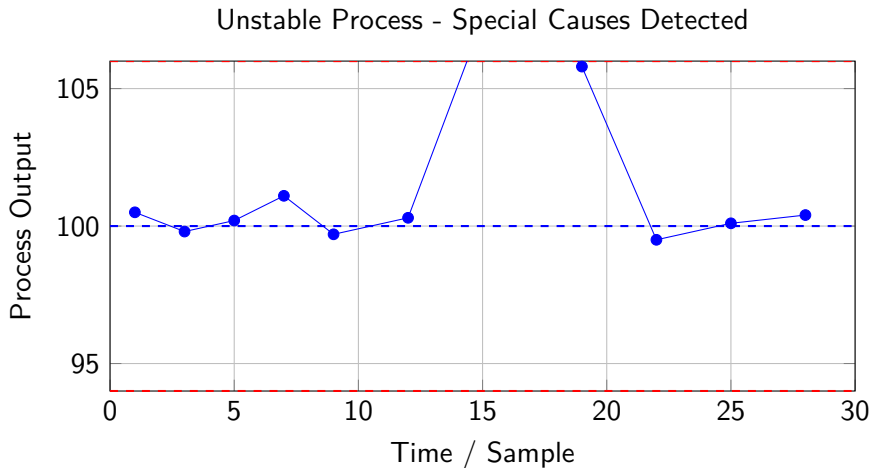
- Assignable, specific causes
- Non-random, unusual events
- Detectable on control charts
- Process is **out of control**
- Can (and should) be eliminated

Visualizing Process Variation

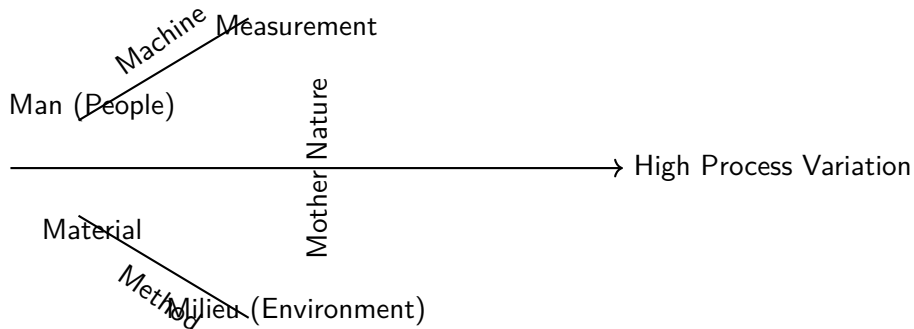
Stable Process - Only Common Causes



Unstable Process – Special Causes Present



Sources of Variation – The 6Ms (Fishbone)



Impact of Variation on Quality

- Increases defects and rework
- Reduces process capability (C_p and C_{pk})
- Leads to customer dissatisfaction
- Increases costs (Cost of Poor Quality)
- Makes prediction and planning difficult
- Hides root causes of problems

Rule of Thumb: Reducing variation is often more important than centering the process.

Strategies to Reduce Variation

- ➊ Identify and remove Special Causes (using Control Charts)
- ➋ Reduce Common Cause Variation through:
 - ▶ Process redesign and standardization
 - ▶ Better training and methods
 - ▶ Improved equipment maintenance
 - ▶ Supplier quality improvement
 - ▶ Environmental control
- ➌ Apply DMAIC / Six Sigma projects
- ➍ Use Big Data for deeper variation analysis

Note: You can always use the 5 Whys technique to dig to the root cause of all problems.

The 5 Whys Technique

- A simple yet powerful root cause analysis tool
- Developed by Taiichi Ohno at Toyota as part of the Toyota Production System (TPS)
- Core Idea: Ask "**Why?**" repeatedly (typically 5 times) to drill down from symptoms to the fundamental/root cause
- Helps move from treating **symptoms** to solving the real **root cause**.

Purpose of 5 Whys in Manufacturing

- Identify the true root cause of defects, downtime, quality issues, or safety incidents
- Prevent recurrence of problems
- Support continuous improvement
- Simple, no statistical software required and can be used on the shop floor
- Complements advanced tools like Fishbone Diagram, Pareto Analysis, and DMAIC

Step-by-Step Process of 5 Whys

- 1 Clearly define the **problem** (be specific)
- 2 Ask "**Why?**" — record the answer
- 3 Ask "**Why?**" again for the previous answer
- 4 Repeat until you reach the **root cause** (usually 5 times, sometimes 3–7)
- 5 Develop **corrective and preventive actions**
- 6 Assign **responsibility** (tagging) and timeline for resolution
- 7 Verify effectiveness after implementation

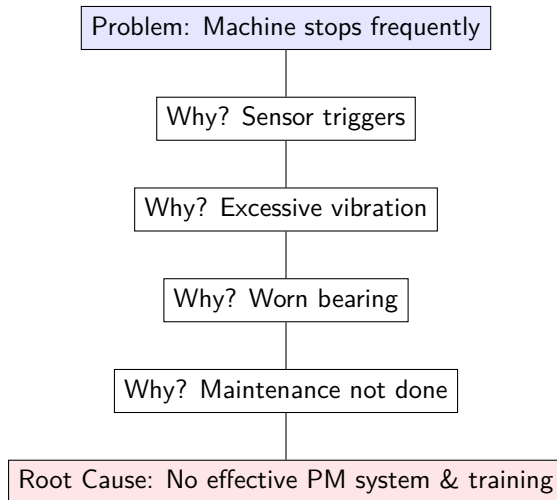
5 Whys – Manufacturing Example

Problem: Machine keeps stopping unexpectedly during production.

- Why 1: The machine overload sensor is triggering.
- Why 2: There is too much vibration on the machine.
- Why 3: The bearing is worn out and causing imbalance.
- Why 4: The preventive maintenance schedule was not followed.
- Why 5: There is no proper maintenance tracking system and operators are not trained on importance of scheduled maintenance.

Root Cause: Lack of effective preventive maintenance system and training.

5 Whys Diagram (Visual)



Tagging & Resolution (Action Plan)

After identifying the root cause:

- **Tag** responsible person/department
- Define specific, measurable corrective actions
- Set deadlines and verification method

Best Practices for Effective 5 Whys

- Involve people who actually work on the process (operators, technicians)
- Use data and facts — avoid assumptions
- Keep asking "Why?" until the cause is actionable and controllable
- Do not stop at human error — go deeper (system/process failure)
- Combine with Fishbone Diagram for complex problems
- Document and share learnings across the organization

Advantages and Limitations

Advantages

- Simple and easy to learn
- Encourages deep thinking
- Low cost, no special tools needed
- Promotes team collaboration
- Fast root cause identification

Limitations

- Can be subjective
- May stop too early (not reaching real root cause)
- Not suitable for very complex problems alone
- Requires experience and domain knowledge

5 Whys in Big Data & Industry

- Use real-time sensor data to identify problems faster
- Combine 5 Whys with machine learning anomaly detection
- Integrate with DMAIC (Analyze phase)
- Digital 5 Whys tools linked with Computerized Maintenance Management System (CMMS) and ERP systems
- Root cause analysis on large-scale production data

Key Takeaways – 5 Whys

- 5 Whys is a powerful, simple tool for root cause analysis in manufacturing
- Focus on **system** issues rather than blaming people
- Always follow up with corrective actions and tagging of responsibility
- Effective when combined with other quality tools
- Essential skill for quality professionals and data analysts working in manufacturing

Variation in Big Data Context

- Sensor drift and calibration issues
- Data pipeline variability (ETL fluctuations)
- Model drift in machine learning systems
- Network latency and cloud resource variation
- Customer behavior variation in service processes

Opportunity: Big Data allows real-time detection and much faster reaction to variation.

Key Takeaways – Process Variation

- All processes have variation and the key is to understand its type
- Distinguish between Common Cause and Special Cause variation
- Control Charts are the primary tool for detecting special causes
- Reducing variation leads to better quality, lower cost, and higher customer satisfaction
- In the Big Data era, variation analysis must be real-time and scalable

Statistical Process Control (SPC) – Core Concepts

- SPC is a statistical method used to monitor, control, and improve processes
- Developed by Walter A. Shewhart in the 1920s
- Uses **control charts** to distinguish between common and special cause variation
- Foundation of modern quality management (Six Sigma, Lean, TQM)
- Key Principle: "Prevention over Detection"
- 7 Basic Quality Tools: Pareto, Fishbone, Histogram, Check Sheet, Scatter Plot, Flowchart, Control Charts

Objectives of Statistical Process Control

- Determine whether a process is in a state of statistical control
- Detect the presence of special causes of variation
- Reduce process variability over time
- Improve process capability and predictability
- Provide a basis for continuous improvement (PDCA / DMAIC)
- Support data-driven decision making

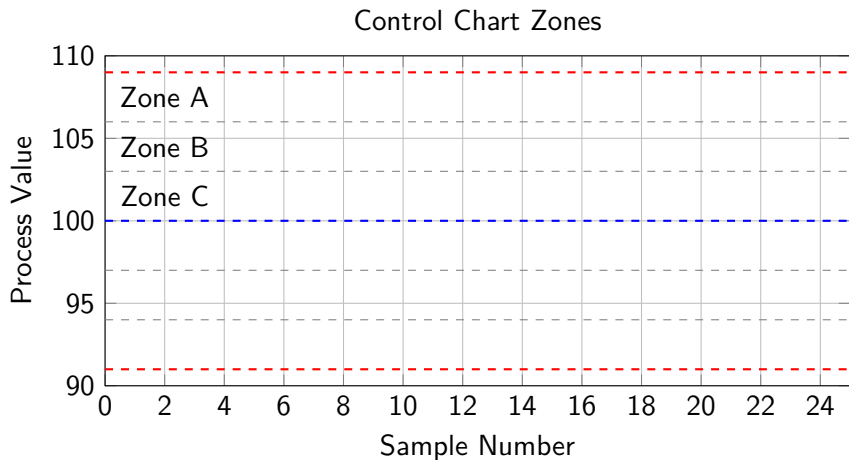
Benefits of SPC

- Early detection of process problems
- Reduced defects and rework
- Lower operating costs
- Improved product consistency and customer satisfaction
- Better process understanding and knowledge
- Foundation for predictive quality in Big Data environments

Control Charts – The Heart of SPC

- Graphical tool that plots process data over time
- Shows whether a process is stable or unstable
- Consists of:
 - ▶ Center Line (CL) — usually the process mean
 - ▶ Upper Control Limit (UCL)
 - ▶ Lower Control Limit (LCL)
- Control limits are typically set at $\pm 3\sigma$ from the centerline

Control Chart Zones and Interpretation



Out-of-Control Signals (Western Electric / Nelson Rules)

- ① Any single point outside the $\pm 3\sigma$ control limits
- ② 2 out of 3 consecutive points in Zone A or beyond
- ③ 4 out of 5 consecutive points in Zone B or beyond
- ④ 8 consecutive points on one side of the centerline
- ⑤ 6 consecutive points steadily increasing or decreasing (trend)
- ⑥ 14 consecutive points alternating up and down
- ⑦ 15 consecutive points in Zone C (stratification)

Control Charts for Variables Data

Variables data = Continuous measurements (weight, length, temperature, time, etc.)

Most Common Charts:

- $\bar{X}$ and R Chart (subgroup size 2–5)
- $\bar{X}$ and S Chart (larger subgroups)
- Individuals & Moving Range (I-MR) Chart ($n=1$)

$\bar{X}$ and R Chart Example

Process: Bottle filling (Target = 500 ml, n=5)

Formulas:

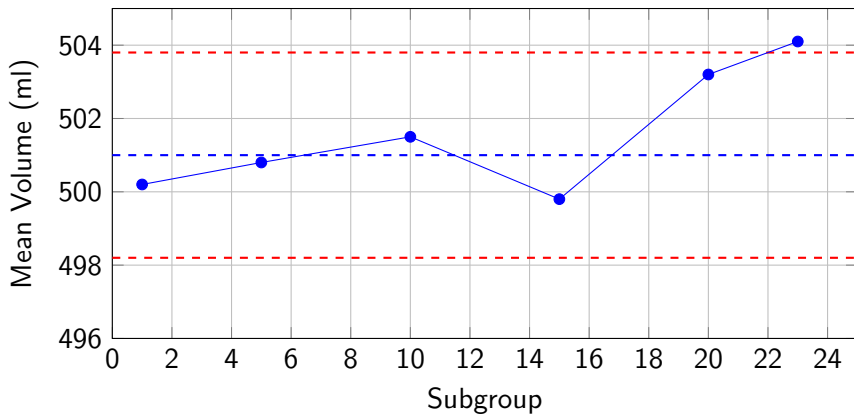
$$UCL_{\bar{X}} = \bar{\bar{X}} + A_2\bar{R}, \quad LCL_{\bar{X}} = \bar{\bar{X}} - A_2\bar{R}$$

$$UCL_R = D_4\bar{R}, \quad LCL_R = D_3\bar{R}$$

(A_2 , D_3 , D_4 are constants from control chart tables)

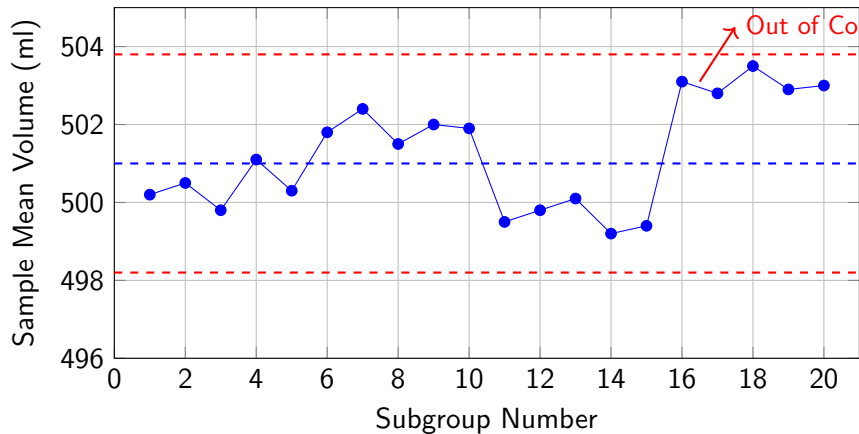
$\bar{X}$ Chart Visualization (Bottle Filling)

$\bar{X}$ Control Chart

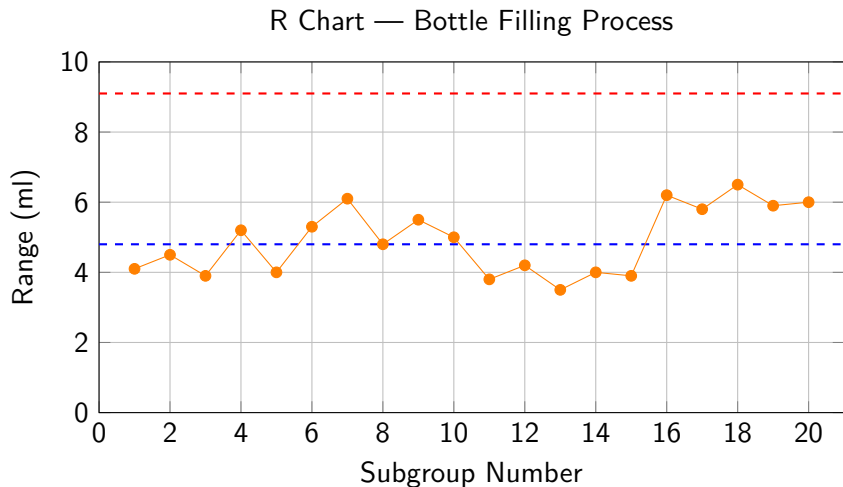


$\bar{X}$ Control Chart — Visualization

$\bar{X}$ Chart — Bottle Filling Process



R Chart (Range)



$\bar{X}$ and R Control Charts

- Most common charts for **variables data** when subgroup size is small ($n = 2$ to 5)
- $\bar{X}$ Chart: Monitors process **mean** (location)
- R Chart: Monitors process **variability** (range)
- Used together to assess process stability

$\bar{X}$ and R Chart – Formulas

Let:

- k = number of subgroups
- n = subgroup size
- $\bar{X}_i$ = mean of i^{th} subgroup
- R_i = range of i^{th} subgroup

$$\bar{\bar{X}} = \frac{\sum \bar{X}_i}{k}, \quad \bar{R} = \frac{\sum R_i}{k}$$

$$UCL_{\bar{X}} = \bar{\bar{X}} + A_2 \bar{R}, \quad LCL_{\bar{X}} = \bar{\bar{X}} - A_2 \bar{R}$$

$$UCL_R = D_4 \bar{R}, \quad LCL_R = D_3 \bar{R}$$

Example: Bottle Filling Process ($\bar{X}$ -R Chart)

Process: Filling bottles with 500 ml target volume. Subgroup size $n = 5$,
Number of subgroups $k = 12$

Subgroup	$\bar{X}$ (ml)	R (ml)
1	501.2	4.1
2	498.8	3.5
3	500.5	5.2
4	502.1	6.8
5	499.7	3.9
6	501.8	4.7
7	500.3	2.8
8	503.4	7.1
9	499.5	4.3
10	501.0	5.0
11	500.8	3.6
12	502.6	6.2

Step-by-Step Calculation

$$\bar{\bar{X}} = 500.98 \text{ ml}, \quad \bar{\bar{R}} = 4.77 \text{ ml}$$

For $n = 5$: $A_2 = 0.577$, $D_3 = 0$, $D_4 = 2.114$

$$UCL_{\bar{X}} = 500.98 + 0.577 \times 4.77 \approx 503.73$$

$$LCL_{\bar{X}} = 500.98 - 0.577 \times 4.77 \approx 498.23$$

$$UCL_R = 2.114 \times 4.77 \approx 10.08, \quad LCL_R = 0$$

Standard Control Chart Constants for Variables Data

Subgroup Size (n)	$\bar{X}$ -R Chart			$\bar{X}$ -S Chart		
	A ₂	D ₃	D ₄	A ₃	B ₃	B ₄
2	1.880	0	3.267	2.659	0	3.267
3	1.023	0	2.574	1.954	0	2.568
4	0.729	0	2.282	1.628	0	2.266
5	0.577	0	2.114	1.427	0	2.089
6	0.483	0	2.004	1.287	0.030	1.970
7	0.419	0.076	1.924	1.182	0.118	1.882
8	0.373	0.136	1.864	1.099	0.185	1.815
9	0.337	0.184	1.816	1.032	0.239	1.761
10	0.308	0.223	1.777	0.975	0.284	1.716
15	0.223	0.348	1.652	0.789	0.428	1.572
20	0.180	0.414	1.586	0.680	0.510	1.490

Most commonly used values highlighted for $n = 4, 5, 6$

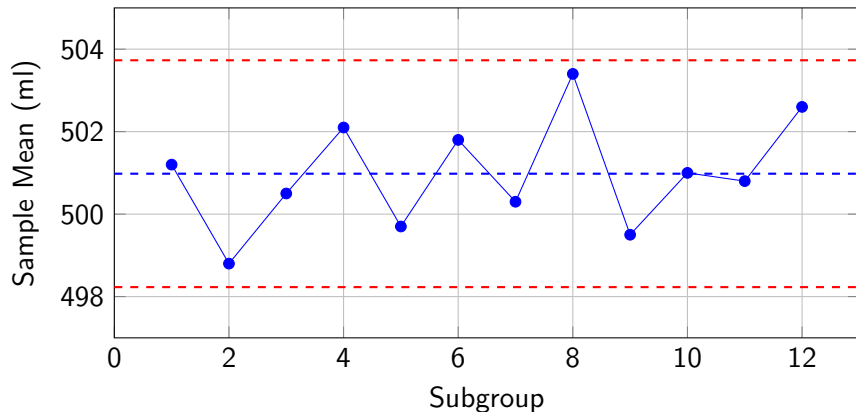
Control Chart Constants Table (Variables)

n	$\bar{X} + R$ Chart			$\bar{X} + S$ Chart		
	A_2	D_3	D_4	A_3	B_3	B_4
2	1.880	0	3.267	2.659	0	3.267
3	1.023	0	2.574	1.954	0	2.568
4	0.729	0	2.282	1.628	0	2.266
5	0.577	0	2.114	1.427	0	2.089
6	0.483	0	2.004	1.287	0.030	1.970
7	0.419	0.076	1.924	1.182	0.118	1.882
8	0.373	0.136	1.864	1.099	0.185	1.815
10	0.308	0.223	1.777	0.975	0.284	1.716

Source: Standard control chart constant tables (ASTM, Montgomery)

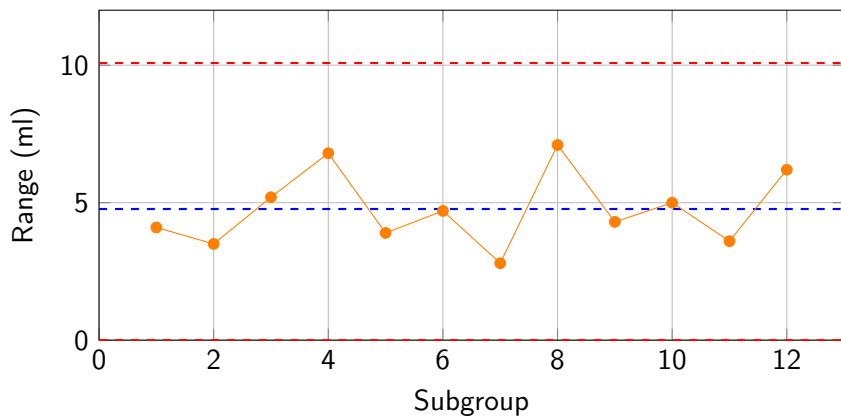
$\bar{X}$ Control Chart

$\bar{X}$ Chart — Bottle Filling Process



R Chart

R Chart — Bottle Filling Process



$\bar{X}$ and S Control Charts

- Preferred when subgroup size $n > 10$
- S Chart uses **standard deviation** instead of range $\rightarrow$ more efficient for larger samples
- Better estimate of process variability

$\bar{X}$ and S Chart – Formulas

$$\bar{\bar{X}} = \frac{\sum \bar{X}_i}{k}, \quad \bar{\bar{S}} = \frac{\sum S_i}{k}$$

$$UCL_{\bar{X}} = \bar{\bar{X}} + A_3 \bar{\bar{S}}, \quad LCL_{\bar{X}} = \bar{\bar{X}} - A_3 \bar{\bar{S}}$$

$$UCL_S = B_4 \bar{\bar{S}}, \quad LCL_S = B_3 \bar{\bar{S}}$$

When to Use R Chart vs S Chart

Situation	Recommended Chart
Subgroup size 2 to 5	$\bar{X}$ -R Chart
Subgroup size 6 to 10	Either (R is still acceptable)
Subgroup size > 10	$\bar{X}$ -S Chart (preferred)
Big Data / very large subgroups	S Chart or EWMA/CUSUM

Key Takeaways

- $\bar{X}$ -R Charts are the most common for small subgroups
- $\bar{X}$ -S Charts are better for larger subgroups
- Always interpret both charts together
- Process must be in statistical control before calculating capability indices
- In Big Data, these charts can be updated in real-time

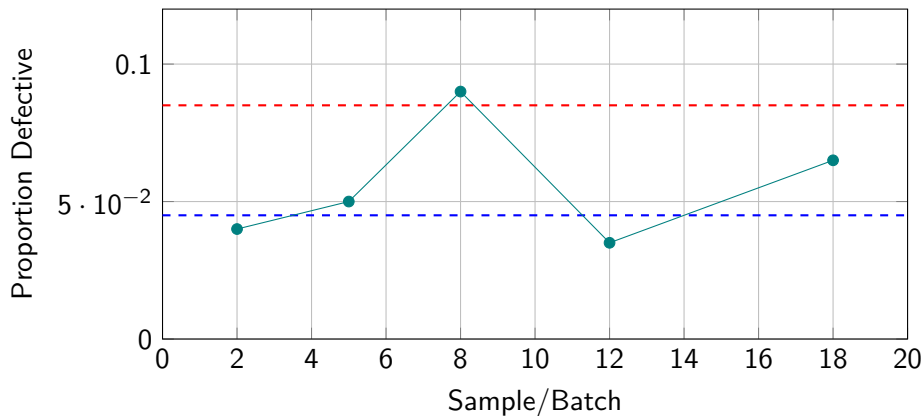
Control Charts for Attribute Data

Attribute data = Count or classification data (pass/fail, defective/non-defective)

- **p-Chart** — Proportion defective (variable sample size)
- **np-Chart** — Number defective (constant sample size)
- **c-Chart** — Number of defects per unit
- **u-Chart** — Defects per unit (variable sample size)

p-Chart Example

p-Chart — Proportion Defective



Control Chart Selection Guide

Data Type	Recommended Chart
Continuous (Variables)	$\bar{X}$ -R, $\bar{X}$ -S, I-MR
Proportion Defective	p-chart, np-chart
Count of Defects	c-chart, u-chart

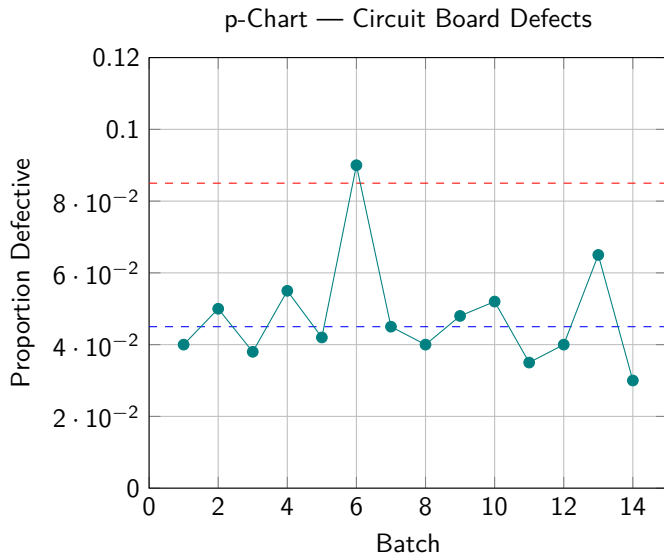
p-Chart (Proportion Nonconforming)

Example: Defective circuit boards

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n_i}}$$

- Used when we monitor the **proportion** of defective items
- Sample size **can vary**
- Based on Binomial distribution
- Most widely used attribute control chart

p-Chart (Proportion Nonconforming)



p-Chart – Formulas

Let:

- n_i = Sample size for i^{th} sample (can vary)
- D_i = Number of defectives in i^{th} sample
- $p_i = \frac{D_i}{n_i}$
- $\bar{p} = \frac{\sum D_i}{\sum n_i}$

Center Line (CL) = $\bar{p}$

$$UCL_i = \bar{p} + 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n_i}}$$

$$LCL_i = \bar{p} - 3\sqrt{\frac{\bar{p}(1 - \bar{p})}{n_i}} \quad (\text{set to 0 if negative})$$

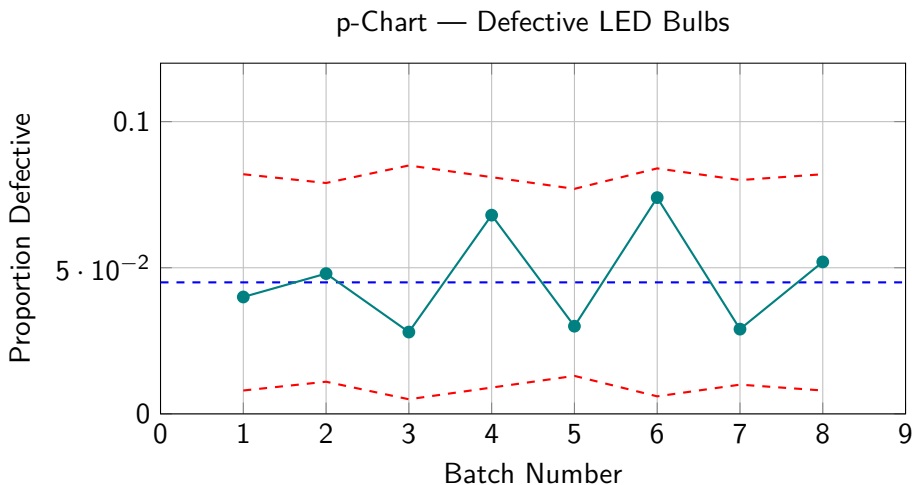
p-Chart – Practical Example

Scenario: Quality inspection of LED bulbs

Batch	Sample Size (n_i)	Defectives (D_i)	Proportion (p_i)
1	200	8	0.040
2	250	12	0.048
3	180	5	0.028
4	220	15	0.068
5	300	9	0.030
6	190	14	0.074
7	240	7	0.029
8	210	11	0.052

$$\bar{p} = 0.045$$

p-Chart Visualization



c-Chart (Count of Nonconformities)

- Used when we count the total number of **defects** (not just defective items)
- Sample size (inspection unit) must be **constant**
- Based on Poisson distribution
- Common in: defects per product, errors per document, scratches per panel, etc.

c-Chart – Formulas

Let:

- c_i = Number of defects in the i^{th} sample
- $\bar{c} = \frac{\sum c_i}{k}$ (average number of defects)

$$\text{Center Line (CL)} = \bar{c}$$

$$\text{UCL} = \bar{c} + 3\sqrt{\bar{c}}$$

$$\text{LCL} = \bar{c} - 3\sqrt{\bar{c}} \quad (\text{set to 0 if negative})$$

c-Chart – Practical Example

Scenario: Number of surface defects on metal sheets (inspection unit = 1 sheet)

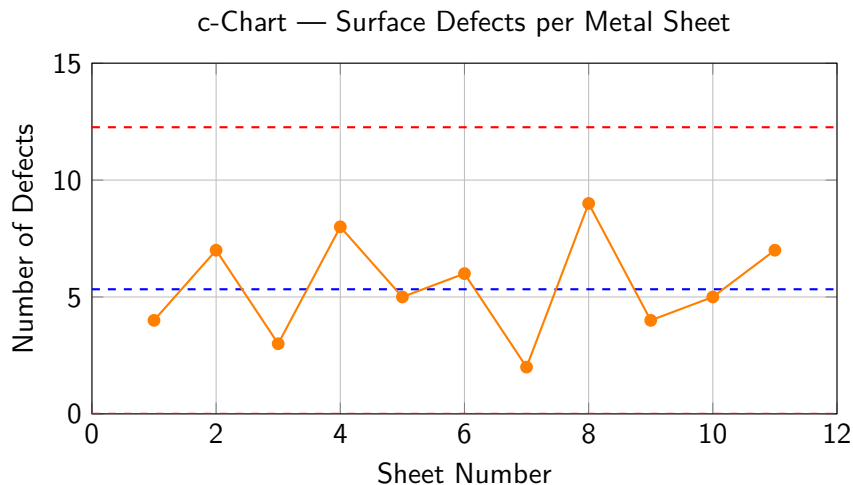
Sheet	1	2	3	4	5	6	7	8	9
Defects	4	7	3	8	5	6	2	9	4

$$\bar{c} = 5.33$$

$$UCL = 5.33 + 3\sqrt{5.33} \approx 12.26$$

$$LCL \approx 0$$

c-Chart Visualization



p-Chart vs c-Chart Comparison

Characteristic	p-Chart	c-Chart
What is measured	Proportion of defectives	Number of defects
Distribution	Binomial	Poisson
Sample size	Can vary	Must be constant
Typical use		

Key Takeaways

- Use **p-Chart** when tracking proportion of defective items (variable sample size)
- Use **c-Chart** when counting total defects in a constant inspection unit
- Both are powerful tools for monitoring attribute data
- In Big Data environments, these charts can be updated in real-time using streaming data

np-Chart (Number of Nonconforming Items)

- Used for **attribute data** when sample size is **constant**
- Monitors the actual **number** of defective items in each sample
- Based on Binomial distribution

np-Chart – Formulas

Let:

- n = Constant sample size
- D_i = Number of defectives in i^{th} sample
- $\bar{np}$ = Average number of defectives = $\frac{\sum D_i}{k}$ (k = number of samples)

$$\bar{p} = \frac{\bar{np}}{n}$$

$$\text{Center Line (CL)} = \bar{np}$$

$$\text{UCL} = \bar{np} + 3\sqrt{\bar{np}(1 - \bar{p})}$$

$$\text{LCL} = \bar{np} - 3\sqrt{\bar{np}(1 - \bar{p})} \quad (\text{set to 0 if negative})$$

np-Chart – Example

Scenario: Inspection of circuit boards. Sample size $n = 100$ boards per batch (constant).

Data (Number of Defectives):

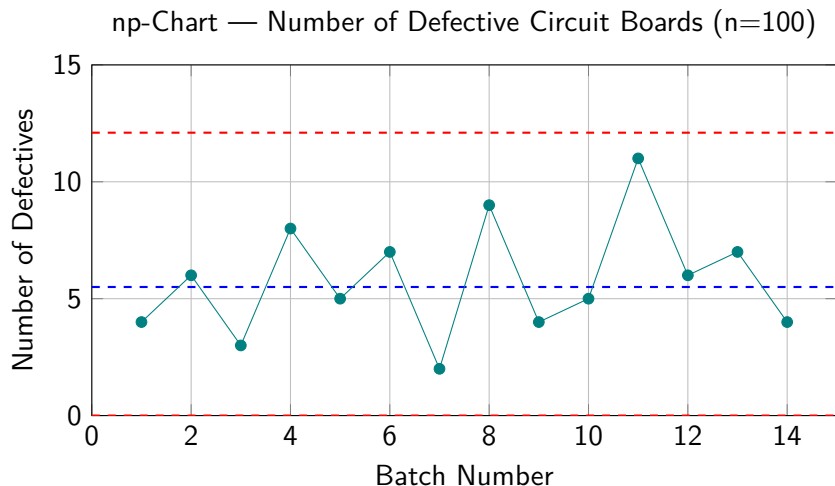
Batch	1	2	3	4	5	6	7	8
Defectives	4	6	3	8	5	7	2	9

$$\bar{n}p = 5.5, \bar{p} = 0.055$$

$$UCL = 5.5 + 3\sqrt{5.5 \times 0.945} \approx 12.1$$

$$LCL = 5.5 - 3\sqrt{5.5 \times 0.945} \approx 0 \quad (\text{negative} \rightarrow 0)$$

np-Chart Visualization



u-Chart (Number of Nonconformities per Unit)

- Used when we count the **number of defects** (not just defectives)
- Sample size (inspection unit) can be ****variable****
- Based on Poisson distribution
- Example: Number of defects per square meter, per 1000 lines of code, per patient day

u-Chart – Formulas

Let:

- c_i = Number of nonconformities (defects) in i^{th} sample
- n_i = Size of i^{th} sample (inspection units)
- $u_i = \frac{c_i}{n_i}$
- $\bar{u} = \frac{\sum c_i}{\sum n_i}$

$$\text{Center Line (CL)} = \bar{u}$$

$$\text{UCL}_i = \bar{u} + 3\sqrt{\frac{\bar{u}}{n_i}}$$

$$\text{LCL}_i = \bar{u} - 3\sqrt{\frac{\bar{u}}{n_i}} \quad (\text{set to 0 if negative})$$

u-Chart – Example

Scenario: Defects in software code review. Inspection unit = 1000 lines of code.

Data:

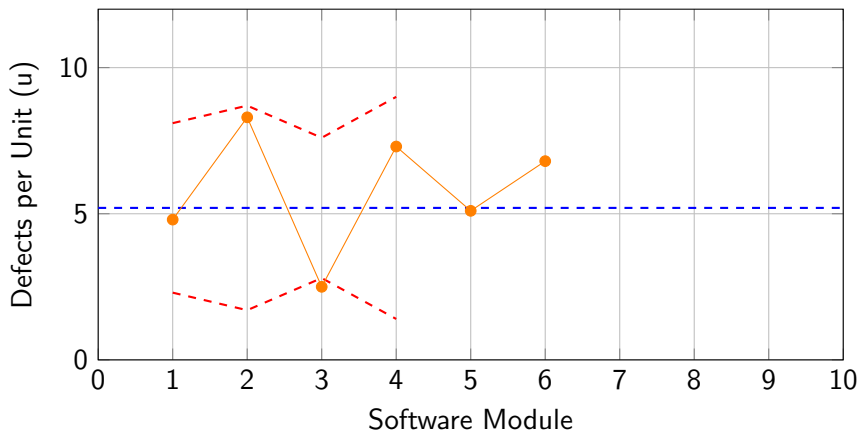
Module	Lines of Code	Defects
1	2500	12
2	1800	15
3	3200	8
4	1500	11

$\bar{u} = 5.2$ defects per 1000 lines

UCL and LCL will vary with sample size n_i .

u-Chart Visualization

u-Chart — Defects per 1000 Lines of Code



Comparison: p / np vs c / u Charts

Chart	What is Counted	Sample Size
p-chart	Proportion of defectives	Variable
np-chart	Number of defectives	Constant
c-chart	Number of defects	Constant
u-chart	Defects per unit	Variable

Key Takeaways – np and u Charts

- Use **np-chart** when sample size is constant and you count defectives
- Use **u-chart** when counting multiple defects per unit and sample size varies
- Both are based on Poisson approximation
- Variable control limits in u-chart due to changing sample size
- Important tools for quality monitoring in manufacturing, software, healthcare, and services

SPC in Big Data Environment

- Streaming control charts using Kafka + Spark / Flink
- Millions of data points processed per minute
- Dynamic / adaptive control limits
- Hybrid systems: Traditional charts + Machine Learning
- Real-time dashboards with automated alerts
- Multivariate and profile monitoring

Key Takeaways – SPC and Control Charts

- SPC helps distinguish common vs special causes
- Control charts are the primary tool for monitoring process stability
- Variables charts for measurement data; Attribute charts for count data
- Stability must be achieved before capability analysis
- In Big Data era, SPC becomes real-time, predictive, and automated

Individuals & Moving Range (I-MR) Chart

Best for: Big Data / sensor data where subgroup size $n=1$ (e.g., temperature, response time)

- I-Chart: Monitors process level
- MR-Chart: Monitors short-term variability

Advantage in Big Data: Ideal for high-frequency streaming data.

Individuals & Moving Range (I-MR) Chart

- Used when subgroup size = 1 ($n=1$).
- Extremely useful in Big Data and continuous processes
- Common applications: sensor readings, temperature, pressure, daily metrics, response time, chemical concentration, server CPU usage, etc.

Why Use I-MR Charts?

- Cannot form traditional subgroups (expensive, destructive, or continuous flow)
- Ideal for high-frequency streaming data in Big Data environments
- Consists of two charts plotted together:
 - ▶ **I-Chart** (Individuals Chart) → Monitors process **level/mean**
 - ▶ **MR-Chart** (Moving Range Chart) → Monitors process variability

I-MR Chart – Formulas

Let x_i = individual measurement at time i

$$\text{Center Line (I-Chart)} = \bar{x} = \frac{\sum x_i}{k}$$

Moving Range: $MR_i = |x_i - x_{i-1}|$

$$\overline{MR} = \frac{\sum MR_i}{k - 1}$$

$$UCL_x = \bar{x} + 2.66\overline{MR}, \quad LCL_x = \bar{x} - 2.66\overline{MR}$$

$$UCL_{MR} = 3.267\overline{MR}, \quad LCL_{MR} = 0$$

I-MR Chart – Practical Example

Scenario: Daily average server response time (in milliseconds) — monitored every hour.

Day	Response Time (x_i)	Moving Range (MR_i)
1	245	-
2	238	7
3	252	14
4	241	11
5	260	19
6	255	5
7	248	7
8	272	24

$$\bar{x} = 251.375 \text{ ms} \quad \overline{MR} = 12.43 \text{ ms}$$

Control Chart Constants for Variables Data

Subgroup	$\bar{X} + R$ Chart			$\bar{X} + S$ Chart			I-MR Chart	
Size (n)	A_2	D_3	D_4	A_3	B_3	B_4	E_2	D_4
2	1.880	0	3.267	2.659	0	3.267	2.660	3.267
3	1.023	0	2.574	1.954	0	2.568	1.772	2.574
4	0.729	0	2.282	1.628	0	2.266	1.457	2.282
5	0.577	0	2.114	1.427	0	2.089	1.290	2.114
6	0.483	0	2.004	1.287	0.030	1.970	1.184	2.004
7	0.419	0.076	1.924	1.182	0.118	1.882	1.109	1.924
8	0.373	0.136	1.864	1.099	0.185	1.815	1.054	1.864
9	0.337	0.184	1.816	1.032	0.239	1.761	1.010	1.816
10	0.308	0.223	1.777	0.975	0.284	1.716	0.975	1.777
15	0.223	0.348	1.652	0.789	0.428	1.572	0.849	1.652
20	0.180	0.414	1.586	0.680	0.510	1.490	0.728	1.586

Note: For I-MR Chart, E_2 is used for Individuals chart limits and D_4 for Moving Range chart limits.

Constants based on standard tables (Montgomery, ASTM).

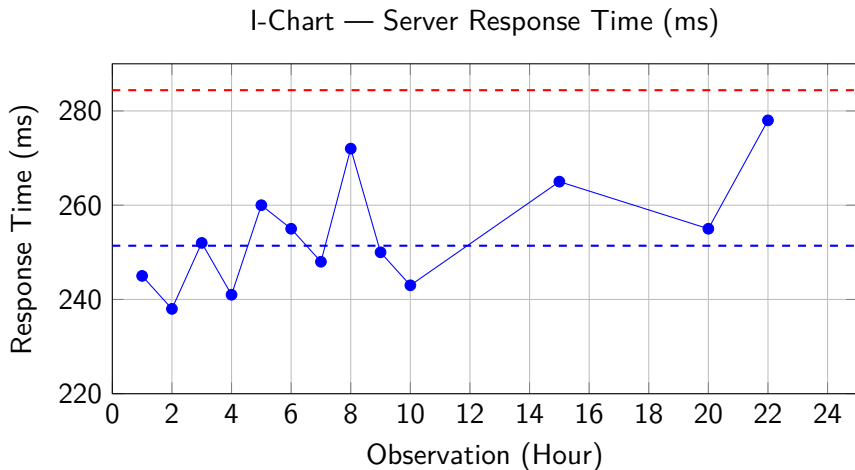
I-MR Chart Constants (n=1)

Constant	Value	Used For
E_2	2.660	Individuals Chart (UCL & LCL)
D_3	0	MR Chart LCL
D_4	3.267	MR Chart UCL
d_2	1.128	Estimation of σ ($\hat{\sigma} = \overline{MR}/d_2$)

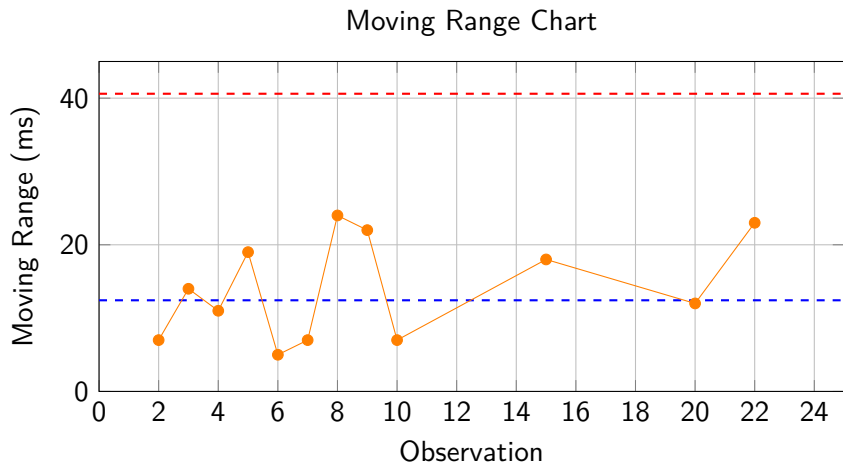
Formulas Reminder:

$$UCL_x = \bar{\bar{x}} + 2.66\overline{MR}, \quad UCL_{MR} = 3.267\overline{MR}$$

I-Chart Visualization



MR-Chart Visualization



I-MR Chart – Big Data Advantages

- Perfect for high-velocity sensor and IoT data
- Can be implemented in real-time using Spark Streaming or Flink
- Moving Range helps detect sudden increases in variability
- Easy to automate with alerts when points go out of control
- Often combined with EWMA or CUSUM for better sensitivity

Key Takeaways – I-MR Charts

- Best choice when subgroup size = 1 (very common in Big Data)
- Two charts: I-Chart (location) + MR-Chart (variation)
- Uses constants 2.66 and 3.267 derived from $d_2 = 1.128$
- Highly valuable for continuous monitoring of temperature, pressure, latency, usage metrics, etc.
- Foundation for real-time quality monitoring in modern data systems

Process Capability Analysis

Key Indices:

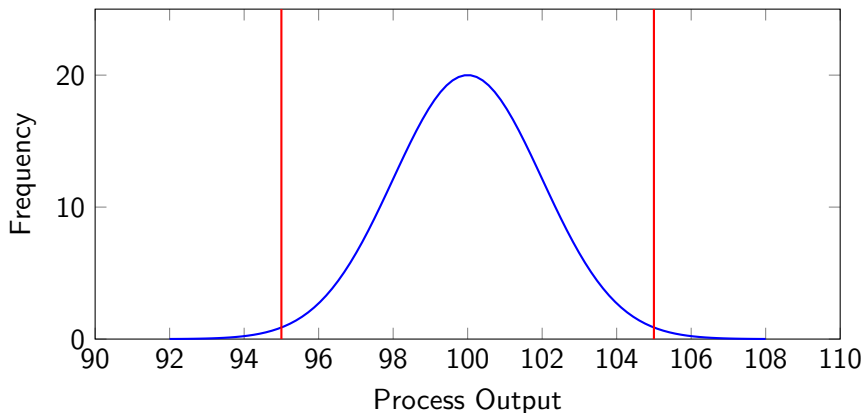
$$C_p = \frac{USL - LSL}{6\sigma} \quad C_{pk} = \min \left(\frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma} \right)$$

Interpretation:

- $C_{pk} > 1.33$: Excellent capability
- $1.00 < C_{pk} \leq 1.33$: Marginal
- $C_{pk} < 1.00$: Not capable

Process Capability Analysis

Capability Histogram



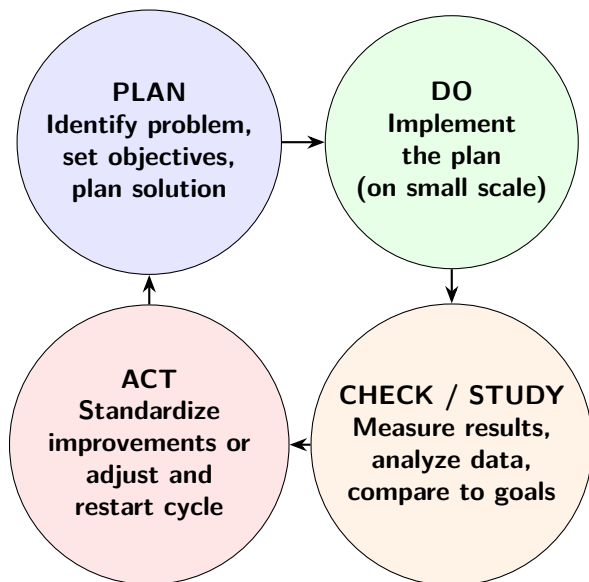
$$C_p = \frac{USL - LSL}{6\sigma}$$

$$C_{pk} = \min \left(\frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma} \right)$$

The Shewhart Cycle

- 1 **Walter A. Shewhart** (1891–1967), the “Father of Statistical Quality Control,” introduced the concept of a cyclical approach to process improvement in the 1930s.
- 2 Originally a **three-step cycle**: - Specification → Production → Inspection
- 3 Later evolved into the **four-step Shewhart Cycle**, widely known today as: - **PDCA** (Plan–Do–Check–Act) or - **PDSA** (Plan–Do–Study–Act) — preferred by Deming
- 4 It is a scientific, iterative method for continuous improvement.

Shewhart Cycle (PDCA / PDSA) Diagram



Shewhart Cycle – Detailed Phases

PLAN

- Identify opportunity or problem
- Collect baseline data
- Set objectives (SMART)
- Develop action plan & prediction

DO

- Execute the plan on a small scale (pilot)
- Document observations

CHECK / STUDY

- Measure and collect data
- Analyze results (Control Charts, Capability, Statistics)
- Compare against objectives
- Identify what worked and what didn't

ACT

- Standardize successful changes
- Adjust and improve the plan
- Restart the cycle for further gains

Shewhart Cycle in Quality Control

- Foundation of ****Continuous Improvement**** (Kaizen)
- Directly supports the ****Control**** phase of DMAIC
- Integrates perfectly with ****Statistical Process Control****
- Used across manufacturing, services, software, and Big Data pipelines

Big Data Application Example: Monitoring and iteratively improving a real-time ETL pipeline using streaming control charts.

Key Takeaways

- Shewhart Cycle (PDCA/PDSA) is the scientific engine of continuous improvement
- Control Charts are essential in the ****Check/Study**** and ****Control**** phases
- Stability → Capability → Continuous Improvement
- Combine classical tools with Big Data for maximum impact

DMAIC Framework – Overview

DMAIC is the core problem-solving methodology of Six Sigma.

- ➊ **Define** — What is the problem?
- ➋ **Measure** — How is the process performing?
- ➌ **Analyze** — What are the root causes?
- ➍ **Improve** — How can we fix it?
- ➎ **Control** — How do we sustain the gains?

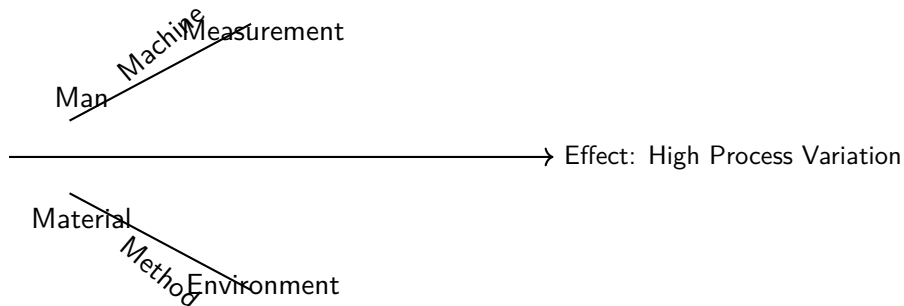
DMAIC – Detailed Phases & Tools

Phase	Key Activities	Tools
Define	Project charter, SIPOC, Voice of Customer	Project Charter, SIPOC
Measure	Data collection, baseline performance	Control Charts, Process Maps
Analyze	Identify root causes	Fishbone, 5 Whys, Pareto
Improve	Implement solutions	DOE, Kaizen, Pilot
Control	Sustain improvements	Control Charts, Control Plans

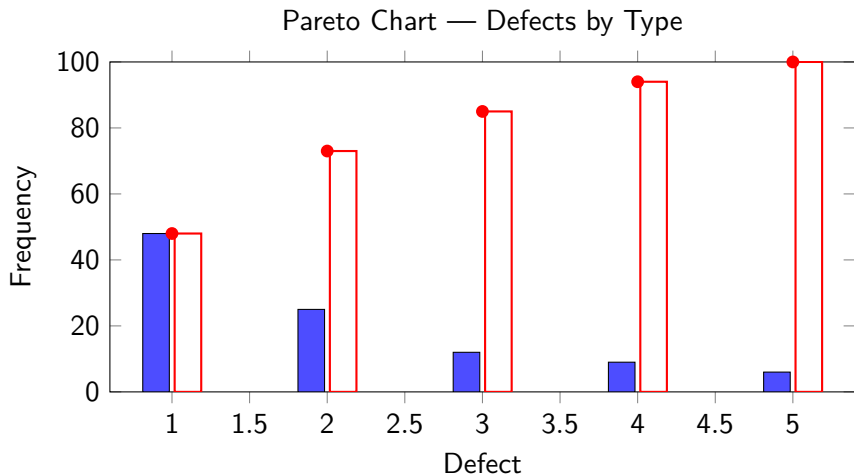
DMAIC Framework

Phase	Tools
Define	SIPOC, Project Charter
Measure	Gauge R&R, Control Charts
Analyze	Fishbone, Pareto
Improve	DOE, Kaizen
Control	SPC, Dashboards

Fishbone (Ishikawa) Diagram



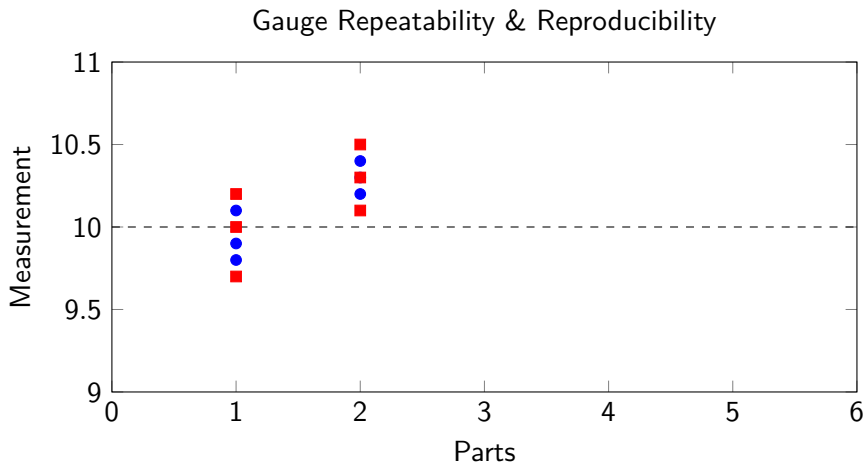
Pareto Chart



Measurement System Analysis (Gauge R&R)

- Critical before trusting data for SPC or Capability Analysis
- Ensures data quality — "Garbage in, Garbage out"
- Evaluates whether the measurement system is adequate
- Primary Method: **Gauge Repeatability & Reproducibility (Gauge R&R)**
- Especially important in Big Data (sensors, APIs, ETL pipelines)
- Determines how much of the total observed variation comes from the measurement system itself

Gauge R&R (Measurement System Analysis)



Goal: Measurement variation « Process variation

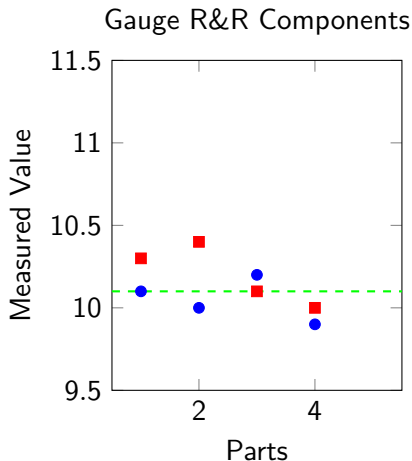
Gauge R&R – Core Concepts

Repeatability

- Variation when **same operator** measures the **same part** multiple times
- Equipment variation (EV)
- "Can the gauge give consistent results?"

Reproducibility

- Variation when **different operators** measure the **same part**
- Appraiser variation (AV)
- "Do different people get the same results?"



Gauge R&R – Sources of Variation

Total Variation = Process Variation + Measurement Variation

$$\text{Total Variation (TV)} = \sqrt{\sigma_{\text{Process}}^2 + \sigma_{\text{Gauge R\&R}}^2}$$

Where:

$$\sigma_{\text{Gauge R\&R}}^2 = \sigma_{\text{Repeatability}}^2 + \sigma_{\text{Reproducibility}}^2$$

Percentage of Study Variation:

$$\%GR\&R = \left(\frac{\sigma_{\text{Gauge R\&R}}}{\sigma_{\text{Total}}} \right) \times 100$$

Gauge R&R – Acceptance Criteria

% GR&R	Interpretation
< 10%	Excellent measurement system
10% – 30%	Acceptable (depends on application)
> 30%	Unacceptable — Improve measurement system first!

- Also evaluate **ndc (Number of Distinct Categories)**: Should be ≥ 5
- Critical before running Process Capability or SPC studies

Gauge R&R Study Design

Standard Crossed Gauge R&R Study:

- 10 parts (representing full process variation)
- 2–3 operators (appraisers)
- 2–3 repeated measurements per part per operator
- Total readings = $10 \times 3 \times 3 = 90$ measurements

Analysis Methods:

- ANOVA (most recommended)
- Average and Range Method (Xbar-R)
- AIAG MSA Manual guidelines

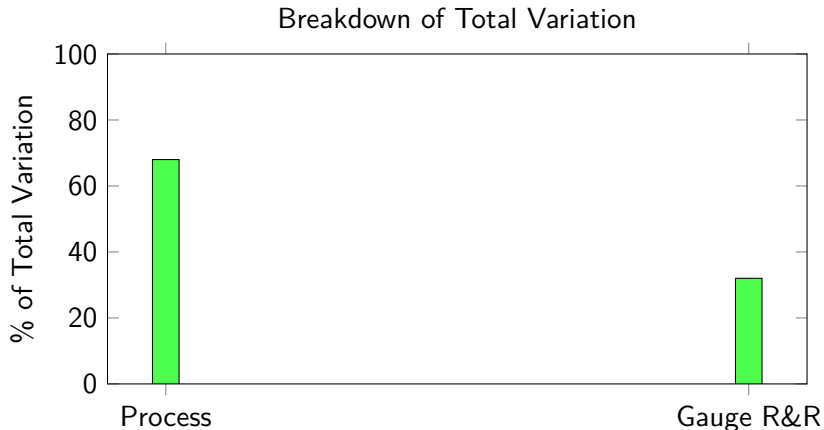
Gauge R&R Example – Big Data Context

Scenario: Automated sensor system measuring temperature in a manufacturing line.

- **Repeatability Issue:** Same sensor gives readings varying by $\pm 1.5^{\circ}\text{C}$ on the same part
- **Reproducibility Issue:** Different data collection scripts produce different averages
- **Result:** $\%GR\&R = 42\% \rightarrow$ **Unacceptable**
- **Action:** Calibrate sensors, standardize data pipelines, implement automated validation

In Big Data environments, poor measurement systems lead to garbage-in-garbage-out analytics.

Gauge R&R Visual Summary

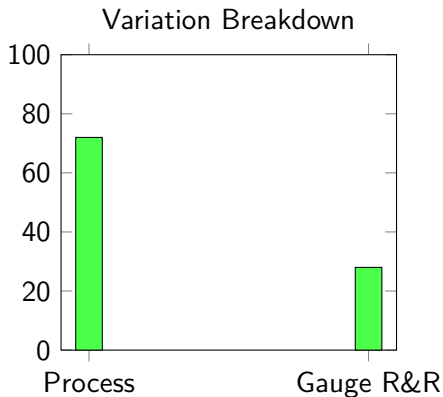


Goal: Maximize Process Variation contribution and minimize Gauge R&R contribution.

Gauge R&R – Components

Repeatability: Same operator, same part

Reproducibility: Different operators, same part



ANOVA Gauge R&R – Study Design

Example Study:

- 4 Parts
- 3 Operators
- 2 Replicates each

ANOVA Gauge R&R – Raw Data

Part	Op1	Op2	Op3
1	10.1, 10.2	10.3, 10.4	10.0, 10.1
2	9.8, 9.9	10.0, 10.1	9.7, 9.8
3	10.5, 10.4	10.6, 10.7	10.3, 10.4
4	9.6, 9.7	9.5, 9.6	9.8, 9.7

ANOVA Table for Gauge R&R

Source	DF	SS	MS	F
Parts	3	2.845	0.948	142.3
Operators	2	0.285	0.142	21.4
Interaction	6	0.095	0.016	2.4
Repeatability	12	0.080	0.0067	—
Total	23	3.305	—	—

Gauge R&R Results

Calculated Results:

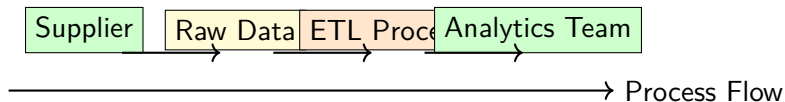
- % Repeatability = 18.2%
- % Reproducibility = 9.8%
- **% Gauge R&R = 28.0%**
- ndc (Number of Distinct Categories) = 4.8

Conclusion: Marginally acceptable. Measurement system needs improvement before using data for control charts or capability studies.

Why Gauge R&R Matters in Big Data Analytics

- Sensors and APIs can have significant measurement error
- Poor data quality invalidates downstream models and control charts
- Must be performed during the **Measure** phase of DMAIC
- Automated Gauge R&R can be implemented using streaming data validation

Value Stream Map (Lean Tool)



Value-Adding Activities vs Non-Value-Adding (Waste)

DMAIC Example 1: Manufacturing (Bottle Filling)

Problem: High variation in filled volume $\rightarrow$ customer complaints.

- **Define:** Reduce volume variation; Target $\sigma < 2$ ml
- **Measure:** Collected data $\rightarrow \bar{X}$ -R charts showed instability (see earlier slide)
- **Analyze:** Fishbone diagram identified filler nozzle wear as root cause
- **Improve:** Replaced nozzles + implemented preventive maintenance $\rightarrow C_{pk}$ improved from 0.85 to 1.45
- **Control:** Implemented real-time $\bar{X}$ -R chart with automated alerts (Spark Streaming)

DMAIC Example 2: Big Data Context (ETL Pipeline)

Problem: High latency and error rate in daily ETL data pipeline.

- **Define:** Reduce ETL processing time by 40% and error rate below 0.5%
- **Measure:** Baseline: Average latency = 85 min, error rate = 2.8%
- **Analyze:**
 - ▶ Pareto chart: 80% errors from 3 data sources
 - ▶ Hypothesis testing: Significant difference in API response times
- **Improve:**
 - ▶ Parallel processing + data validation rules
 - ▶ Result: Latency reduced to 42 min, error rate to 0.4%
- **Control:** Real-time monitoring dashboard with p-chart and anomaly detection model

Key Takeaways

- Use visual tools (Control Charts, SIPOC, Fishbone, Pareto) for better understanding
- Shewhart Cycle + DMAIC = Powerful improvement engine
- Measurement System (Gauge R&R) is critical
- Big Data turns traditional SQC into real-time, predictive quality systems

Acceptance Sampling – Introduction

- Statistical method used to decide whether to accept or reject a lot (batch) of products
- Applied to incoming raw materials, finished goods, or outsourced components
- Not a substitute for process control — it is a form of ****inspection****
- Widely used when 100% inspection is too costly or destructive

Key Concepts in Acceptance Sampling

- **Lot** — A batch of products produced under similar conditions
- **Sample** — Randomly selected items from the lot
- **Acceptance Number (c)** — Maximum number of defectives allowed in sample
- **Producer's Risk (α)** — Probability of rejecting a good lot (Type I error)
- **Consumer's Risk (β)** — Probability of accepting a bad lot (Type II error)
- **AQL** — Acceptable Quality Limit (producer's quality level)
- **LTPD** — Lot Tolerance Percent Defective (consumer's quality level)

Types of Acceptance Sampling Plans

- ① **Single Sampling Plan** — One sample is taken
- ② **Double Sampling Plan** — May take a second sample if first is inconclusive
- ③ **Multiple Sampling Plan**
- ④ **Sequential Sampling Plan** — Items inspected one by one until decision is made

Most Common: Single and Double Sampling Plans

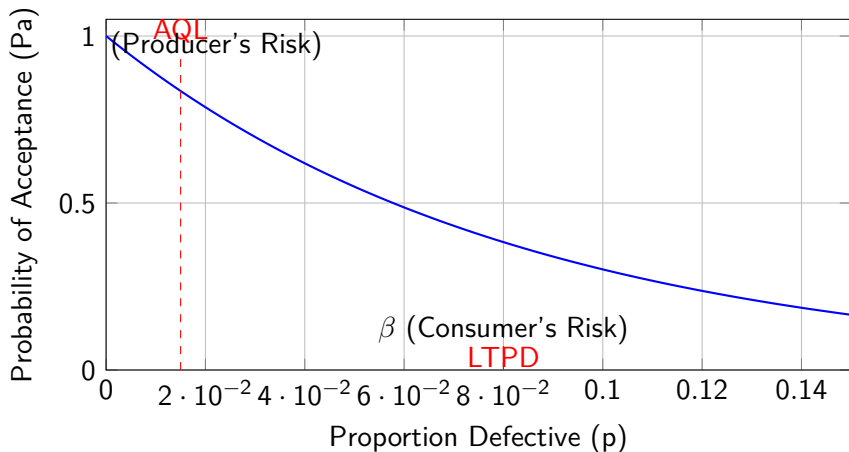
Operating Characteristic (OC) Curve

The OC Curve shows the **probability of accepting a lot** as a function of the actual quality level (proportion defective).

- Ideal OC Curve: Vertical line at AQL
- Real OC Curve: Shows discrimination power of the sampling plan

OC Curve – Visualization

Operating Characteristic (OC) Curve



Single Sampling Plan – Example

Parameters:

- Lot Size (N) = 2000
- Sample Size (n) = 125
- Acceptance Number (c) = 3
- AQL = 1.0%
- LTPD = 5.0%

Decision Rule:

- If number of defectives in sample $\leq 3 \rightarrow$ Accept the lot
- If number of defectives $> 3 \rightarrow$ Reject the lot

Advantages and Disadvantages of Acceptance Sampling

Advantages

- Less expensive than 100% inspection
- Less damage to products
- Useful for destructive testing
- Provides clear decision rules

Disadvantages

- Risk of accepting bad lots or rejecting good lots
- Does not improve the process
- Should not replace SPC
- Requires careful plan design

Six Sigma – Introduction

- Data-driven methodology aimed at reducing process variation
- Goal: Achieve **3.4 Defects Per Million Opportunities (DPMO)**
- Originated at Motorola (1986) by Bill Smith
- Popularized by General Electric under Jack Welch
- Combines statistical tools with business strategy

Six Sigma Performance Levels

Sigma Level	DPMO	Process Yield
6σ	3.4	99.99966%
5σ	233	99.977%
4σ	6,210	99.38%
3σ	66,807	93.32%
2σ	308,537	69.15%

DMAIC – The Six Sigma Roadmap

- ➊ **Define** — Define problem, goals, customer requirements (SIPOC, Voice of Customer)
- ➋ **Measure** — Collect baseline data, validate measurement system (Gauge R&R)
- ➌ **Analyze** — Identify root causes (Fishbone, Pareto, Hypothesis Testing, Regression)
- ➍ **Improve** — Develop and implement solutions (DOE, Kaizen, Pilot)
- ➎ **Control** — Sustain gains (Control Charts, Control Plans, Dashboards)

Six Sigma Roles & Certification

- **Champions** — Senior managers who sponsor projects
- **Master Black Belts** — Experts who train and mentor
- **Black Belts** — Full-time project leaders
- **Green Belts** — Part-time project team members
- **Yellow Belts** — Basic awareness

Lean Six Sigma

- Combines **Six Sigma** (variation reduction) with **Lean** (waste elimination)
- Focus on both quality and speed
- Key Lean Tools: 5S, Value Stream Mapping, Kanban, Poka-Yoke, Just-in-Time
- 8 Wastes (Muda): Transportation, Inventory, Motion, Waiting, Overproduction, Overprocessing, Defects, Unused Talent

Six Sigma in Big Data Analytics Context

- Improve quality of data pipelines and ETL processes
- Reduce model drift and prediction errors
- Optimize A/B testing and experimentation
- Improve reliability of real-time analytics systems
- Reduce Cost of Poor Data Quality
- Apply DMAIC to machine learning model performance

Key Takeaways

- Acceptance Sampling is for lot acceptance decisions with known risks
- OC Curve helps evaluate the performance of a sampling plan
- Six Sigma aims for near-perfect quality through variation reduction
- DMAIC is the structured problem-solving framework
- Lean Six Sigma combines speed and quality
- These tools are highly relevant for Big Data quality improvement projects

Total Quality Management (TQM) – Introduction

- A comprehensive management philosophy focused on continuous improvement of quality
- Involves **everyone** in the organization — from top management to frontline employees
- Shifts focus from product inspection to **process improvement** and **customer satisfaction**
- Foundation for modern frameworks like Six Sigma, Lean, and ISO 9001

Key Contributors to TQM

Expert	Major Contribution
W. Edwards Deming	14 Points for Management, PDCA Cycle, System of
Joseph Juran	Quality Trilogy (Planning, Control, Improvement), Pa
Philip Crosby	"Quality is Free", Zero Defects, Cost of Poor Quality
Kaoru Ishikawa	Fishbone Diagram, 7 QC Tools, Company-wide Qual
Genichi Taguchi	Robust Design, Loss Function

Core Principles of TQM

- **Customer Focus** — Internal & external customers
- **Total Employee Involvement** — Empowerment and teamwork
- **Process-Centered Approach** — Manage processes, not just outcomes
- **Integrated System** — All departments work toward common goals
- **Strategic & Systematic Approach**
- **Continual Improvement** (Kaizen)
- **Fact-Based Decision Making** — Use data and statistical tools
- **Effective Communication**
- **Leadership Commitment**

Deming's 14 Points for Management (Summary)

- Create constancy of purpose
- Adopt the new philosophy
- Cease dependence on inspection
- End awarding business on price tag alone
- Improve constantly the system
- Institute training on the job
- Institute leadership
- Drive out fear
- Break down barriers between departments
- Eliminate slogans and targets
- Eliminate numerical quotas
- Remove barriers to pride of workmanship
- Institute education and self-improvement
- Put everybody to work on the transformation

Cost of Poor Quality (COPQ)

Four Categories (Crosby & Juran)

- **Prevention Costs** — Training, process planning, quality audits
- **Appraisal Costs** — Inspection, testing, calibration
- **Internal Failure Costs** — Rework, scrap, downtime
- **External Failure Costs** — Warranty, returns, lost goodwill, reputation damage

Key Message: Prevention is much cheaper than failure.

TQM Implementation Framework

- 1 Top Management Commitment
- 2 Customer Focus & Voice of Customer (VoC)
- 3 Employee Training and Empowerment
- 4 Process Management & Standardization
- 5 Continuous Improvement (PDCA / Kaizen)
- 6 Fact-Based Management (SPC, Data Analytics)
- 7 Recognition and Reward Systems
- 8 Supplier Partnership

TQM in Big Data Analytics Context

- Data Quality Management as core part of TQM
- Quality of algorithms and machine learning models
- Real-time quality monitoring of data pipelines
- Continuous improvement of analytical processes
- Building quality culture in data science teams
- Integration with DevOps (DataOps / MLOps)
- Reducing "Data Debt" and technical debt

Benefits of TQM

- Higher customer satisfaction and loyalty
- Reduced costs and waste
- Improved process efficiency and productivity
- Better employee morale and engagement
- Enhanced competitive advantage
- Foundation for Industry 4.0 and data-driven organizations

Big Data Applications in SQC – Why It Matters

- Traditional SQC was designed for small samples and batch processes
- Modern industries generate massive volumes of high-velocity data (IoT, sensors, logs)
- Big Data enables **real-time**, **predictive**, and **automated** quality control
- Combines classical statistical tools with modern data engineering and machine learning
- Creates significant competitive advantage through reduced defects, lower costs, and higher customer satisfaction

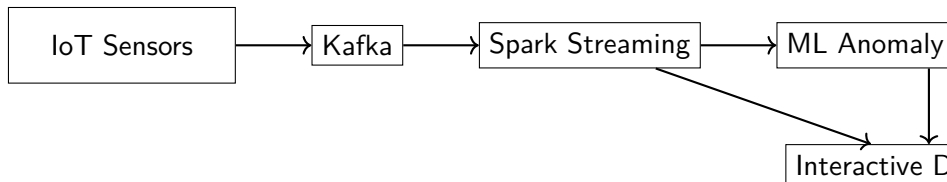
Traditional SQC vs Big Data-Enabled SQC

Traditional SQC	Big Data SQC
Small sample sizes	Millions of data points per minute
Periodic (batch) analysis	Real-time streaming analysis
Manual chart interpretation	Automated anomaly detection
Reactive	Predictive & Prescriptive
Limited to manufacturing	Applies to manufacturing, services, software, I

Real-Time Statistical Process Control

- Traditional control charts updated every few hours/days
- Big Data enables ****sub-second**** updates
- Technologies: Apache Kafka, Spark Streaming, Flink, Apache Storm
- Dynamic control limits that adapt to data volume and velocity
- Automated alerts and root cause analysis

Architecture for Real-Time SPC



Advanced Control Charts for Big Data

- **EWMA** (Exponentially Weighted Moving Average) — Sensitive to small shifts
- **CUSUM** (Cumulative Sum) — Excellent for detecting small persistent shifts
- Adaptive control limits using large historical data
- Multivariate Control Charts (Hotelling's T^2) for correlated variables
- Profile Monitoring for functional data (e.g., waveforms)

Machine Learning + SPC for Anomaly Detection

- Traditional 3-sigma rules + Modern ML algorithms
- Unsupervised techniques:
 - ▶ Isolation Forest
 - ▶ Autoencoders
 - ▶ One-Class SVM
- Hybrid approach: Use control charts for interpretability + ML for detection power
- Real-time scoring on streaming data

Large-Scale Process Capability Analysis

- Traditional: Small samples, normality assumption
- Big Data:
 - ▶ Non-parametric capability indices
 - ▶ Rolling capability analysis on sliding windows
 - ▶ Handling massive datasets with Spark / Dask
 - ▶ Capability analysis across multiple dimensions (multivariate)
- Predictive Capability using time-series forecasting

Predictive Quality Analytics

- Move from reactive to ****predictive**** quality
- Predict defects before they occur using:
 - ▶ Time-series forecasting
 - ▶ Survival analysis
 - ▶ Deep Learning on sensor patterns
- Digital Twin + Quality Prediction
- Remaining Useful Life (RUL) prediction for equipment

Case Study 1: Smart Manufacturing

Scenario: Automotive assembly line with 10,000+ sensors

- Real-time $\bar{X}$ and EWMA charts on Spark
- Anomaly detection using Isolation Forest
- Predictive maintenance reduced unplanned downtime by 35%
- Automated quality alerts sent to operators via mobile

Case Study 2: Data Pipeline Quality (Big Data ETL)

Problem: High error rate in daily ETL pipeline

- Implemented real-time p-chart on data validation results
- Used ML to detect anomalous data patterns
- Reduced error rate from 2.8% to 0.3%
- Built self-healing data quality rules

Challenges in Big Data SQC

- Volume, Velocity, Variety, Veracity
- High-dimensional data
- Non-stationary processes
- Data quality of the quality data itself
- Interpretability of black-box ML models
- Computational cost and scalability

Opportunities for Big Data Analysts

- Develop end-to-end quality monitoring platforms
- Build hybrid SPC + ML systems
- Create interactive quality dashboards (Power BI, Grafana, Superset)
- Design automated closed-loop quality control systems
- Research in areas like Deep SPC, Quantum SPC, and Edge Computing quality

Key Takeaways

- Big Data transforms SQC from periodic inspection to real-time intelligent monitoring
- Integration of classical tools (Control Charts, Capability) with modern technologies is powerful
- Predictive and prescriptive quality is the future
- Data quality itself must be managed as a core process
- MSc Big Data graduates are ideally positioned to lead quality transformation initiatives